Author Contributions Checklist Form

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This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

☐ This paper does not involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

The data analysis uses the Early Childhood Longitudinal Study, Kindergarten Class of 1998-99 (ECLS-K), Kindergarten-Eighth Grade public-use child file, released by the National Center for Education Statistics. The derived file `data_analysis/education.data.csv` (21,409 children, 21 variables) is an extract of that file. From it, the analysis forms 3 x 3 tables of primary nonparental care before kindergarten (center-based care, relative care, no nonparental care) by fall-kindergarten mathematics proficiency level (1-3) for Black girls (N = 94) and Black boys (N = 112) in low-income, single-parent families. All simulation data are generated by the code.

Availability

☒ Data **are** publicly available.
☐ Data **cannot be made** publicly available.

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

☒ Data are available online at: the NCES DataLab online codebook, https://nces.ed.gov/datalab/onlinecodebook (ECLS-K:1998-99 Kindergarten-Eighth Grade Public-Use File; child-level file `ECLSK_98_99_K8_CHILD_v1_0.csv`). Study landing page: https://nces.ed.gov/ecls/kindergarten.asp. No registration or data use agreement is required.

☒ Data are available as part of the paper's supplementary material. The derived extract `data_analysis/education.data.csv` is included, so the data analysis can be reproduced without downloading the ECLS-K file.

☐ Data are publicly available by request, following the process described here:

☐ Data are available through some other mechanism, described here:

Non-publicly available data

Not applicable.

Description

File format(s)

- ☒ CSV or other plain text.
- ☐ Software-specific binary format (.Rda, Python pickle, etc.):
- ☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
- ☐ Other (please specify):

Data dictionary

- ☒ Provided by authors in the following file(s): `README.txt` (section DATA AVAILABILITY lists the variables used, their ECLS-K names, and the recoding) and the comments at the beginning of `data_analysis/education_data_analysis.R`, which document each variable and the extraction of `education.data.csv` from the ECLS-K file.
- ☐ Data file(s) is (are) self-describing (e.g., netCDF, HDF5, Zarr files)
- ☒ Available at the following URL: ECLS-K Kindergarten-Eighth Grade user's manuals, https://nces.ed.gov/ecls/data/eclsk_k8_manual_part1.pdf and https://nces.ed.gov/ecls/data/eclsk_k8_manual_part2.pdf (variable definitions and coding).

Additional Information (optional)

Column names and codes in `education.data.csv` are those of the ECLS-K public-use file (version 1.0). The commented block at the beginning of `education_data_analysis.R` re-creates `education.data.csv` from `ECLSK_98_99_K8_CHILD_v1_0.csv` by keeping 21 variables; running it on the downloaded file reproduces the provided extract exactly.

Part 2: Code

Abstract

The `library/` folder implements the methods of the paper in R and C++ (`sensitivityIxJ.R`, `sensitivityIxJ.cpp`, `SIS_methods.cpp`): the exact worst-case p-value of ordinal, sign-score, and general count-based tests under the generic bias model, the sampling-based estimators, and the normal approximation based on the exact moments. The remaining folders hold the scripts that reproduce the results: the ECLS-K data analysis, the power and size simulations, the timing comparison of the kernel representation, the corner-solution counterexample, the running-time and memory benchmarks, and the verification scripts, together with the previous release of the engine kept for comparison.

Description

Code format(s)

- ☒ Script files
  - ☒ R
  - ☐ Python
  - ☐ Julia
  - ☐ MATLAB
  - ☒ Other: C++ source files, compiled at run time by `Rcpp::sourceCpp()`
- ☐ Package
  - ☐ R
  - ☐ Python
  - ☐ Julia
  - ☐ MATLAB toolbox
  - ☐ Other:
- ☐ Reproducible report

- ☐ Quarto or R Markdown
- ☐ Jupyter notebook
- ☐ Other:

☒ Shell script

☒ Other (please specify): an HTCondor submit description (`power_simulation/submit.txt`) with its parameter grid (`params.txt`)

Supporting software requirements

Primary software used R version 4.4.3 or later (the reported results were obtained with R version 4.6.1), with a C++ compiler for Rcpp. Optional, only to re-derive the profile-likelihood scores in the data analysis: Python 3.13.3 with Gurobi Optimizer 12.0.2 (a Gurobi license is required). The recorded scores are entered in the script, and this step is off by default.

Libraries and dependencies used by the code R packages, with the versions used for the reported results: Rcpp 1.1.2 (all scripts), multicool 1.0.1, BiasedUrn 2.0.12, arrangements 1.1.10, sensitivitymv 1.4.4, dplyr 1.2.1, ggplot2 4.0.3, patchwork 1.3.2. Optional, for re-deriving the profile-likelihood scores only: reticulate 1.47.0, with the Python packages gurobipy and numpy. The role of each package is given in `README.txt`, section REQUIREMENTS, which also lists these versions.

Supporting system/hardware requirements (optional)

All results except the power simulations were obtained on Posit Cloud (Linux) with 8 CPUs and 4 GB RAM; no GPU is used. The power simulations were run on an HTCondor cluster, one job per (seed, gamma) pair, each job in an Apptainer/Singularity container with R (4.4 or later), Rcpp, and a C++ compiler, requesting 4 CPUs and 20 GB of memory; the same scripts also run on a single machine, as described in the README.

Parallelization used

- ☐ No parallel code used
- ☐ Multi-core parallelization on a single machine/node
  - Number of cores used:
- ☒ Multi-machine/multi-node parallelization
  - Number of nodes and cores used: The power simulations are distributed with HTCondor: 110 independent jobs per test (10 seeds x 11 values of gamma; five tests in all), each on one node and single-threaded (4 cores requested per job). All other scripts are single-threaded.

License

- ☒ MIT License (default)
- ☐ BSD
- ☐ GPL v3.0
- ☐ Creative Commons
- ☐ Other: (please specify)

Additional information (optional)

The materials are also maintained in a GitHub repository with the same folder structure. The repository is private during review and will be made public at publication; its location and the release tag corresponding to the published version are given in the Location section below.

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

- ☒ Any numbers provided in text in the paper
- ☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))
- ☐ All tables and figures in the paper
- ☒ Selected tables and figures in the paper, as explained and justified below:

The workflow reproduces every table and figure that reports a computed quantity: the sensitivity-analysis tables of the data analysis (main text and Supplementary Material), the power tables, the size-control figures, the convergence figures of the sampling-based estimators, the computation-time tables, and the running-time and memory benchmarks. Three figures are illustrations (the introductory example of an I x J table, the collapsed and cross-cut tables, and the integer-to-dummy mapping) and involve no computation. The power-curve figures of the main text and the Supplementary Material plot the values of the power tables against gamma; each value is the total number of rejections written by the power scripts divided by 1,000. The Cross-Cut column of those tables (Fisher's exact test on the corner cells of the 3 x 3 table: treatment levels 1 and 3, outcome levels 1 and 3) is obtained from either 2 x 2 script by replacing the line that builds `observed.table` with `observed.table = matrix(c(n11, n13, n31, n33), nrow = 2, byrow = TRUE)`, as in the cross-cut tables of `data_analysis/education_data_analysis.R`; see README.txt, HOW TO REPRODUCE THE RESULTS, item 3. Computation times are reproduced up to the variation expected between machines; the chi-square benchmark for 3 x 4 tables is reported for N up to 120, and the entries marked X in that table were not computed.

Workflow

Location

The workflow is available:

- ☒ As part of the paper's supplementary material.
- ☒ In this Git repository: private during review; the public location and release tag will be added at publication.
- ☐ Other (please specify):

Format(s)

- ☐ Single master code file
- ☒ Wrapper (shell) script(s)
- ☐ Self-contained Quarto, R Markdown file, Jupyter notebook, or other literate programming approach
- ☒ Text file (e.g., a readme-style file) that documents workflow
- ☐ Makefile
- ☒ Other (more detail in *Instructions* below)

Instructions

Full instructions are in `README.txt`, section HOW TO REPRODUCE THE RESULTS (items 1-9). In short: set the R working directory to the top-level folder of the repository (the one containing `README.txt`); every script begins by compiling and sourcing the library from `library/`, so no installation is needed. Each result is reproduced by sourcing one script by its path relative to the top-level folder: `data_analysis/education_data_analysis.R` (data analysis), `power_simulation/Kernel_Advantage_Simulation.R` (computation-time comparison and convergence figures), `power_simulation/size_control.R` (size control), `power_simulation/Corner_Solution_Counter_Example.R` (counterexample), and the scripts in `feasibility/` (benchmarks). The power simulations are run either on an HTCondor cluster (`condor_submit`

`power_simulation/submit.txt`, then the shell script `command_generating_summary.txt` collects the rejection counts) or cell by cell on a single machine with `Rscript`, for which the README gives the command and its expected output. The scripts in `development_check/` verify the code and should all report PASS.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

- ☐ < 1 minute
- ☐ 1-10 minutes
- ☐ 10-60 minutes
- ☐ 1-8 hours
- ☒ > 8 hours
- ☐ Not feasible to run on a desktop machine, as described here:

The total is dominated by the two chi-square benchmarks over the full candidate set (about 4 hours for 3 x 3 tables and about 18 hours for 3 x 4 tables with N up to 120). Everything else runs in about 4 hours on a single machine: the data analysis, the counterexample, and the verification scripts take seconds to minutes; the size-control simulation about an hour; the computation-time comparison several minutes per table for the six tables of its first part, plus the sampling experiments of its second part; the ordinal benchmarks a few minutes; and the power simulations one to two minutes per (test, gamma) cell, 55 cells in all. Running times for each benchmark cell are reported in Section G of the Supplementary Material. All times were measured on Posit Cloud with 8 CPUs and 4 GB RAM and depend on the CPU and RAM available.

Additional information (optional)

The main functions in `library/sensitivityIxJ.R` are documented in the file (roxygen-style headers with arguments, returned values, and examples), and `README.txt` describes each function. The expected p-values of the data analysis are recorded as comments beside the corresponding calls in `education_data_analysis.R`. The scripts in `development_check/` show how the engine is checked against independent results (the multivariate hypergeometric distribution and `fisher.test()` at gamma = 0, total probability mass one, exact moments) and against the previous release in `legacy/`.

Notes (optional)

Random seeds are fixed in all scripts; the exact p-values involve no randomness and are reproduced to the printed precision on any platform. The `legacy/` folder contains the previous release of the engine and is used only by the verification script that compares the two engines. Author information was removed from the materials for review and will be restored in the public release.